

7.

```
siri_chandana@LAPTOP-MMV x + v
siri_chandana@LAPTOP-MMV:~$ ls
Certifications Practical input.txt practical3.c processconcepts4.c processid sjf.c week1-5.c week1-9.c
Ex1 ex.c output.txt practical4.c processconcepts5.c processid.c week1-2.c week1-6.c week1.c
OS1.txt.save fcfs.c practical1.c priority.c processconcepts6.c roundrobin.c week1-3.c week1-7.c
OS_Lab filecopy practical2.c processconcepts3.c processconcepts1.c shell week1-4.c week1-8.c
siri_chandana@LAPTOP-MMV:~$ cd Practical
siri_chandana@LAPTOP-MMV:~/Practical$ ls
Week-1
siri_chandana@LAPTOP-MMV:~/Practical$ mkdir practical7
siri_chandana@LAPTOP-MMV:~/Practical$ cd practical7
siri_chandana@LAPTOP-MMV:~/Practical/practical7$ pwd
/home/siri_chandana/Practical/practical7
siri_chandana@LAPTOP-MMV:~/Practical/practical7$
```

```
GNU nano 8.7.1 practical7.c *
#include <stdio.h>
#include <stdlib.h>

int global = 100;
static int s = 50;

int main()
{
    int stack = 10;
    int *heap = (int *)malloc(sizeof(int));

    printf("Code Segment : %p\n", (void *)main);
    printf("Global Segment : %p\n", (void *)&global);
    printf("Static Segment : %p\n", (void *)&s);
    printf("Heap Segment : %p\n", (void *)heap);
    printf("Stack Segment : %p\n", (void *)&stack);

    free(heap);

    return 0;
}
```

```
siri_chandana@LAPTOP-MMV x + v
siri_chandana@LAPTOP-MMV:~$ ls
Certifications Practical input.txt practical3.c processconcepts4.c processid sjf.c week1-5.c week1-9.c
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OS_Lab filecopy practical2.c processconcepts3.c processconcepts1.c shell week1-4.c week1-8.c
siri_chandana@LAPTOP-MMV:~$ cd Practical
siri_chandana@LAPTOP-MMV:~/Practical$ ls
Week-1
siri_chandana@LAPTOP-MMV:~/Practical$ mkdir practical7
siri_chandana@LAPTOP-MMV:~/Practical$ cd practical7
siri_chandana@LAPTOP-MMV:~/Practical/practical7$ pwd
/home/siri_chandana/Practical/practical7
siri_chandana@LAPTOP-MMV:~/Practical/practical7$ nano practical7.c
siri_chandana@LAPTOP-MMV:~/Practical/practical7$ gcc practical7.c -o practical7
siri_chandana@LAPTOP-MMV:~/Practical/practical7$ ./practical7
Code Segment : 0x5b323496e1a9
Global Segment : 0x5b3234971010
Static Segment : 0x5b3234971014
Heap Segment : 0x5b3254cd2010
Stack Segment : 0x7ffff413fccc
siri_chandana@LAPTOP-MMV:~/Practical/practical7$ |
```

8a.

```
siri_chandana@LAPTOP-MMV x + v
GNU nano 8.7.1 practical8a.c *
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int *a, *b;

    // malloc()
    a = (int *)malloc(5 * sizeof(int));

    // calloc()
    b = (int *)calloc(5, sizeof(int));

    // realloc()
    a = (int *)realloc(a, 10 * sizeof(int));

    // free()
    free(a);
    free(b);

    return 0;
}
```

```

siri_chandana@LAPTOP-MMVUDEP:~/Practical/practical8a$ nano practical8a.c
siri_chandana@LAPTOP-MMVUDEP:~/Practical/practical8a$ gcc practical8a.c -o practical8a
siri_chandana@LAPTOP-MMVUDEP:~/Practical/practical8a$ ./practical8a
siri_chandana@LAPTOP-MMVUDEP:~/Practical/practical8a$ valgrind --version
valgrind-3.26.0
siri_chandana@LAPTOP-MMVUDEP:~/Practical/practical8a$ valgrind --leak-check=full ./practical8a
==818== Memcheck, a memory error detector
==818== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==818== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==818== Command: ./practical8a
==818==
==818==
==818== HEAP SUMMARY:
==818==     in use at exit: 0 bytes in 0 blocks
==818==   total heap usage: 3 allocs, 3 frees, 80 bytes allocated
==818==
==818== All heap blocks were freed -- no leaks are possible
==818==
==818== For lists of detected and suppressed errors, rerun with: -s
==818== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
siri_chandana@LAPTOP-MMVUDEP:~/Practical/practical8a$ |

```

8b.

```

GNU nano 8.7.1 practical8b.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int x = 100;

    if (fork() == 0)
    {
        printf("Child Before = %d\n", x);

        x = 500;

        printf("Child After = %d\n", x);
    }
    else
    {
        sleep(2);
        printf("Parent = %d\n", x);
    }

    return 0;
}

siri_chandana@LAPTOP-MMVUDEP:~/Practical/practical8a$ valgrind --version
valgrind-3.26.0
siri_chandana@LAPTOP-MMVUDEP:~/Practical/practical8a$ valgrind --leak-check=full ./practical8a
==818== Memcheck, a memory error detector
==818== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==818== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==818== Command: ./practical8a
==818==
==818==
==818== HEAP SUMMARY:
==818==     in use at exit: 0 bytes in 0 blocks
==818==   total heap usage: 3 allocs, 3 frees, 80 bytes allocated
==818==
==818== All heap blocks were freed -- no leaks are possible
==818==
==818== For lists of detected and suppressed errors, rerun with: -s
==818== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
siri_chandana@LAPTOP-MMVUDEP:~/Practical/practical8a$ cd ~/Practical
siri_chandana@LAPTOP-MMVUDEP:~/Practical$ mkdir practical8b
siri_chandana@LAPTOP-MMVUDEP:~/Practical$ cd practical8b
siri_chandana@LAPTOP-MMVUDEP:~/Practical/practical8b$ nano practical8b.c
siri_chandana@LAPTOP-MMVUDEP:~/Practical/practical8b$ gcc practical8b.c -o practical8b
siri_chandana@LAPTOP-MMVUDEP:~/Practical/practical8b$ ./practical8b
Child Before = 100
Child After = 500
Parent = 100
siri_chandana@LAPTOP-MMVUDEP:~/Practical/practical8b$ |

```

